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**Answer all questions. Calculators and mobile phones are NOT allowed.**

1. [10 pts.] Let  $A$  and  $B$  be  $n \times n$  matrices such that  $AB = -BA$ . Show that if  $n$  is odd, then  $AB$  is singular.

2. [10 pts.] Show that for all  $x \in \mathbb{R}$ , 
$$\begin{vmatrix} 1 & 0 & x & x \\ 0 & 1 & 1 & 1 \\ 1 & x & 0 & x \\ 1 & x & x & x \end{vmatrix} \geq 0.$$

3. [10 pts.] Let  $U$  and  $V$  be vectors in  $\mathbb{R}^n$  such that  $\|U\| = 2$  and  $\|V\| = 3$ . Find the maximum possible value of  $\|3U - 2V\|$ .

4. [10 pts.] Let  $A$  be an  $n \times n$  matrix. Show that if  $A$  is nonsingular matrix then  $\text{adj}A$  is nonsingular.

5. [10 pts.] Given that  $(\text{adj}A)^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ 3 & 4 & 2 & 0 \\ 5 & 6 & 7 & 2 \end{bmatrix}$ , find  $A$ .

6. [10 pts.] Let  $U_1 = (1, 0, 1, 2)$ ,  $U_2 = (2, 0, 2, 3)$  and  $U_3 = (3, 0, 2, 2)$  be vectors in  $\mathbb{R}^4$ . Determine whether the vector  $X = (1, 2, 0, 4)$  can be written as a linear combination of  $U_1, U_2$  and  $U_3$ .

7. [10 + 10 = 20 pts.] Determine whether the given sets are subspaces of  $\mathbb{R}^3$ . Justify your answer.

(a)  $W_1 = \{(a - b + 2c, 2a + 3b - c, a + 5b + c) : a, b, c \in \mathbb{R}\}$ .

(b)  $W_2 = \{X \in \mathbb{R}^3 : \|X\| \leq 1\}$ .

8. [4 × 5 = 20 pts.] Answer each of the following as True/ False. Justify your answer.

(a) If  $A$  is a  $3 \times 3$  nonsingular matrix, then  $|-A^2 \text{adj}(A^{-1})| = -1$ .

(b) If  $A$  is a  $5 \times 5$  matrix such that  $|A| = -3$ , then there exists a matrix  $B$  such that  $\text{adj}B = A$ .

(c) If  $A$  and  $B$  are  $n \times n$  matrices, then  $|A + B^T| = |A^T + B|$ .

(d) If  $X$  and  $Y$  are vectors in  $\mathbb{R}^n$ , then  $\text{span}\{X + Y, X - Y\} \subseteq \text{span}\{X, Y\}$ .

1. [10 pts.] Since  $n$  is odd, if  $AB = -BA \Rightarrow |AB| = |-AB| = (-1)^n |AB| = -|AB|$ . Therefore,  $|AB| = 0$ . Hence,  $AB$  is singular.

2. [10 pts.] 
$$\begin{vmatrix} 1 & 0 & x & x \\ 0 & 1 & 1 & 1 \\ 1 & x & 0 & x \\ 1 & x & x & x \end{vmatrix} = \begin{vmatrix} 1 & 0 & x & x \\ 0 & 1 & 1 & 1 \\ 0 & x & -x & 0 \\ 0 & x & 0 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ x & -x & 0 \\ x & 0 & 0 \end{vmatrix} = \begin{vmatrix} x & -x \\ x & 0 \end{vmatrix} = x^2 \geq 0.$$

3. [10 pts.]  $\|3U - 2V\| \leq \|3U\| + \|-2V\| = 3\|U\| + |-2|\|V\| = 3\|U\| + 2\|V\| = 3(2) + 2(3) = 12$ .

4. [10 pts.]  $A \operatorname{adj} A = |A|I \Rightarrow |A \operatorname{adj} A| = ||A|I|$ . Therefore,  $|A| |\operatorname{adj} A| = |A|^n$ . Since  $|A| \neq 0$ ,  $|\operatorname{adj} A| = |A|^{n-1} \neq 0$ . Thus,  $\operatorname{adj} A$  is nonsingular.

5. [10 pts.] Given that  $(\operatorname{adj} A)^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ 3 & 4 & 2 & 0 \\ 5 & 6 & 7 & 2 \end{bmatrix}$ , find  $A$ .

We have  $|(\operatorname{adj} A)^{-1}| = 8$ . Therefore,  $|\operatorname{adj} A| = \frac{1}{8}$ . Since  $|\operatorname{adj} A| = |A|^3$ , then  $|A| = \frac{1}{2}$ .

Now  $(\operatorname{adj} A)^{-1} = \frac{1}{|A|} A$ . Hence,  $A = |A|(\operatorname{adj} A)^{-1} = \frac{1}{2}(\operatorname{adj} A)^{-1}$ .

6. [10 pts.] Form  $c_1 U_1 + c_2 U_2 + c_3 U_3 = X$ . This leads to a linear system whose augmented

matrix is: 
$$\left[ \begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 0 & 0 & 2 \\ 1 & 2 & 2 & 0 \\ 2 & 3 & 2 & 4 \end{array} \right].$$
 Clearly the system has no solution.

7. [10 + 10 = 20 pts.] (a)  $W_1 = \{(a - b + 2c, 2a + 3b - c, a + 5b + c) : a, b, c \in \mathbb{R}\} = \{a(1, 2, 1) + b(-1, 3, 5) + c(2, -1, 1) : a, b, c \in \mathbb{R}\} = \operatorname{span}\{(1, 2, 1), (-1, 3, 5), (2, -1, 1)\}$ . Therefore,  $W_1$  is a subspace of  $\mathbb{R}^3$ .

(b)  $W_2 = \{X \in \mathbb{R}^3 : \|X\| \leq 1\}$ .

Let  $X = (1, 0, 0)$  and  $c = 2$ . It is clear that  $X \in W_2$ ,  $\|X\| = 1$ . Now  $cX = 2(1, 0, 0) = (2, 0, 0) \notin W_2$ ,  $\|cX\| = 2$ . Therefore,  $W_2$  is not a subspace of  $\mathbb{R}^3$ .

8. [ $4 \times 5 = 20$  pts.] True/False.

(a) If  $A$  is a  $3 \times 3$  nonsingular matrix, then  $|-A^2 \operatorname{adj}(A^{-1})| = -1$ .

TRUE.  $|-A^2 \operatorname{adj}(A^{-1})| = |-A^2 \frac{1}{|A|} A| = -|A|^3 \frac{1}{|A|^3} = -1$ .

(b) If  $A$  is a  $5 \times 5$  matrix such that  $|A| = -3$ , then there exists a matrix  $B$  such that  $\operatorname{adj} B = A$ .

FALSE. If true,  $|\operatorname{adj} B| = |B|^4 = |A| = -3$ , a contradiction.

(c) If  $A$  and  $B$  are  $n \times n$  matrices, then  $|A + B^T| = |A^T + B|$ .

TRUE.  $|A + B^T| = |(A + B^T)^T| = |A^T + B|$ .

(d) If  $X$  and  $Y$  are vectors in  $\mathbb{R}^n$ , then  $\operatorname{span}\{X + Y, X - Y\} \subseteq \operatorname{span}\{X, Y\}$ .

TRUE.  $X + Y$  and  $X - Y$  are linear combinations of  $X$  and  $Y$ .